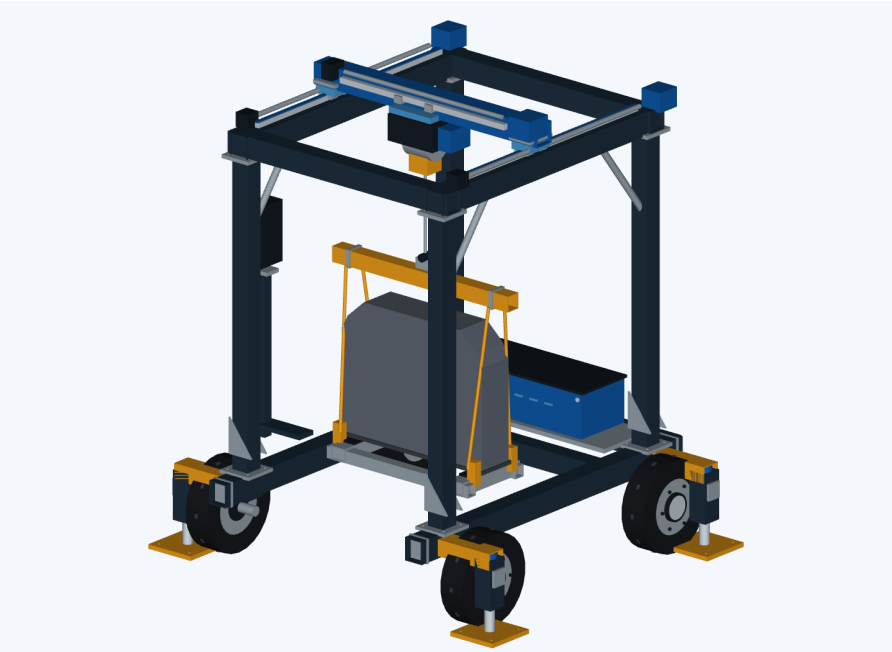
INDEPENDENT ENGINEERING PROJECT / DESIGN AND STABILITY

RoverSetter

Mobile monument handling through prototype development, CAD, and mechanical analysis.



Current RoverSetter CAD assembly. Design documentation, not a field photograph.

Problem and development

A monument installation through my business took about three to four hours and damaged a stone. I developed RoverSetter to investigate a compact mobile system for maneuvering and positioning cemetery monuments. The long-term load target was 2,000 lb, which the prototypes did not demonstrate.

Iteration	Mechanical work
Prototype 1	Initial mechanical proof of concept.
Prototype 2	Revised mechanism and layout after the first approach proved unsatisfactory.
Prototype 3	Motorized chassis with a forward hoist/positioning concept.

Prototype 3 observation

With an approximately 70 lb stone suspended ahead of the front wheel-contact line on hard ground, both rear wheels lifted nearly together while the chassis was stationary. The machine rocked and settled; pitching became more pronounced when the drive motors accelerated it forward.

STATIC ANALYSIS / FINDINGS

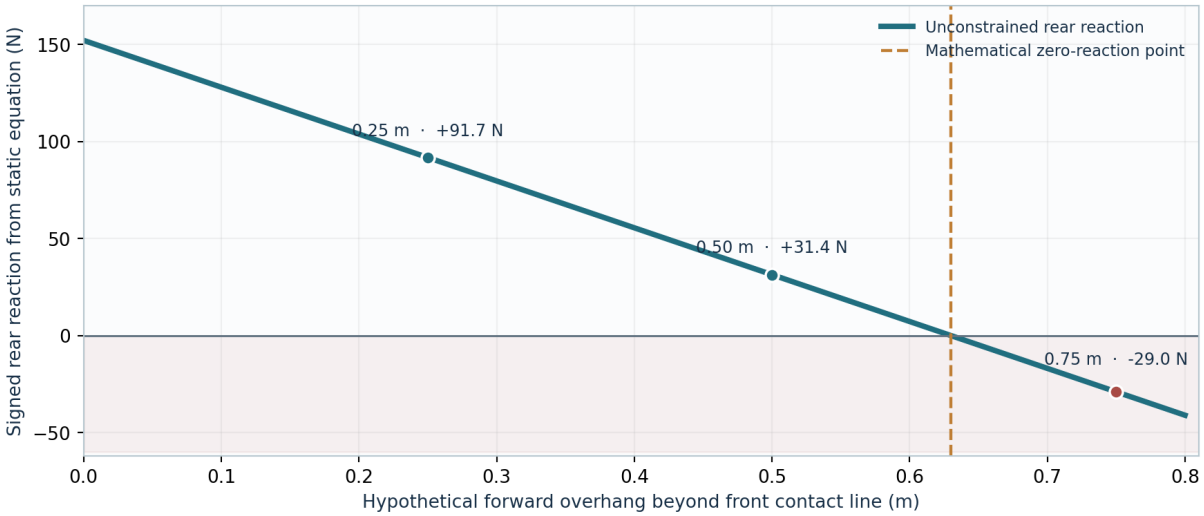
Investigating load position

I returned to the moment balance after realizing the positioning system could move the suspended load forward relative to the machine. A fixed-position calculation would miss changes in the total rear support reaction across the intended travel.

N_rear = (M g L - m g d) / B

M is machine mass, L its center-of-mass arm behind the front wheels, m the suspended load mass, d the forward overhang, and B the support spacing. For fixed geometry and masses, dN_rear/dd = -mg/B.

Forward overhang lowers theoretical rear support



MODEL ONLY: M=40 kg, L=0.50 m, B=1.29 m (assumed); m≈31.75 kg (recalled estimate). Negative values are INFEASIBLE contacts.

Example model only: M = 40 kg, L = 0.50 m, B = 1.29 m; m ≈ 31.75 kg. These inputs are not historical machine measurements.

Assumed overhang	Calculated signed rear reaction
0.25 m	+91.7 N
0.50 m	+31.4 N
0.75 m	-29.0 N: four-contact static equilibrium infeasible

Conclusion

Increasing load overhang reduces the calculated rear support reaction in the static model. This helps explain the sensitivity of the machine to load location; it does not establish the cause of the stronger motor-startup pitching. That response needs dynamic measurements, including chassis motion, load motion, and changing wheel contacts.

Project status: three physical builds; follow-up CAD and Python analysis; seven unit tests. The full-size handling target and machine stability have not been experimentally validated.